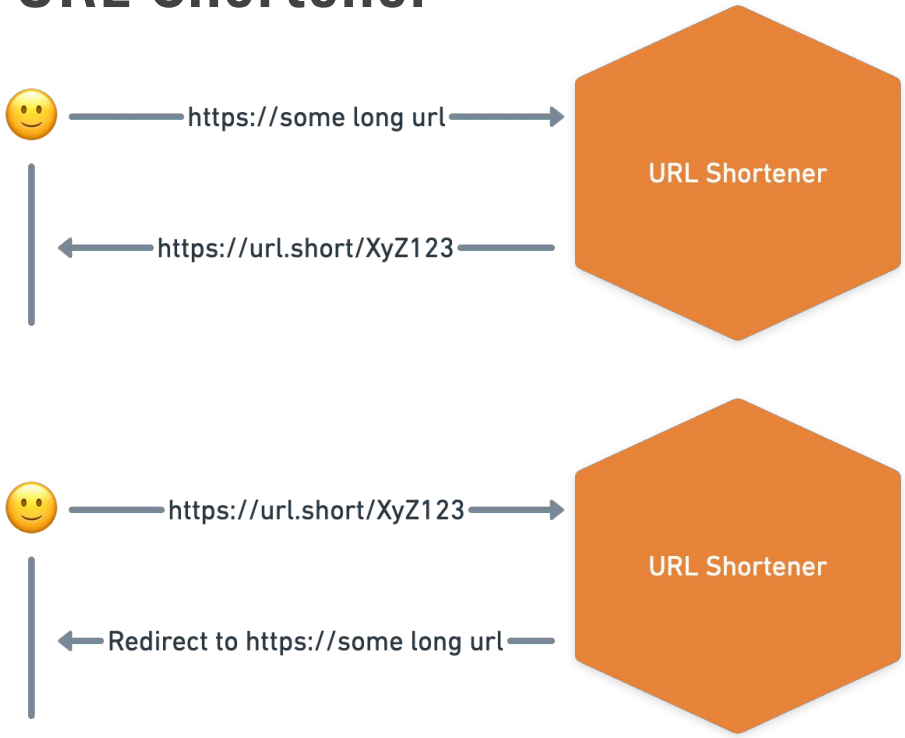
Design URL Shortener

Problem Introduction

Benefits of short URLs

- Easier to type
- Don’t take much space when you share them

How it works - URL Shortener



Questions

- How many new URLs will be added per month?
- For how long the URLs should be kept?
- Total number of URLs?

Answers

- How many new URLs will be added per month?
  - 500M
- For how long the URLs should be kept?
  - 5 years
- Total number of URLs?
  - 500M * 12 * 5 = 30 Billion URLs

Short identifies

Requirements

- As short as possible
- Random
- Human readable

UUID?

0-9 + a-f = 16 characters

16^32 => (2^4)^32 => 2^128

Too long to type!

❌ Short

✅ Random

✅ Human readable

MD5?

Size: 2^128, same as UUID 👍

```
substring(Base58(Md5(URL)), 8) 🤔
```

Not random!

❌Short

❌ Random

✅Human readable

Base62

All English letters = 52 + 10 numbers = 62

UUID = 32 characters

Base62(UUID) = 22 characters

62^5 = 916 Million

62^6 = 56 Billion 👍

62^7 = 3 Trillion

62^8 = 281 Trillion

✅Short

✅ Random

❌Human readable

Base58

All characters - zero, capital i, capital o and lower case L

58^6 = 38 Billion 👍

58^7 = 2 Trillion

58^8 = 128 Trillion

✅Short

✅ Random

✅ Human readable

Base58

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1. substring(Base58(UUID), 6)

2. For 1 to 8
    a. Random(0, 57)
    b. append

Basic Design

URL Shortener

Writes per second

500M URLs shortened per month => ~600M/30

20M per day => ~24M/24

1M per hour => ~1.2M/60

20K per minute => ~18K / 60

300 URLs per second

X10 peak? => 3K URLs

Storage requirements

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30 billion URLs

Each URL 100 bytes on average =>

3 Trillion bytes =>

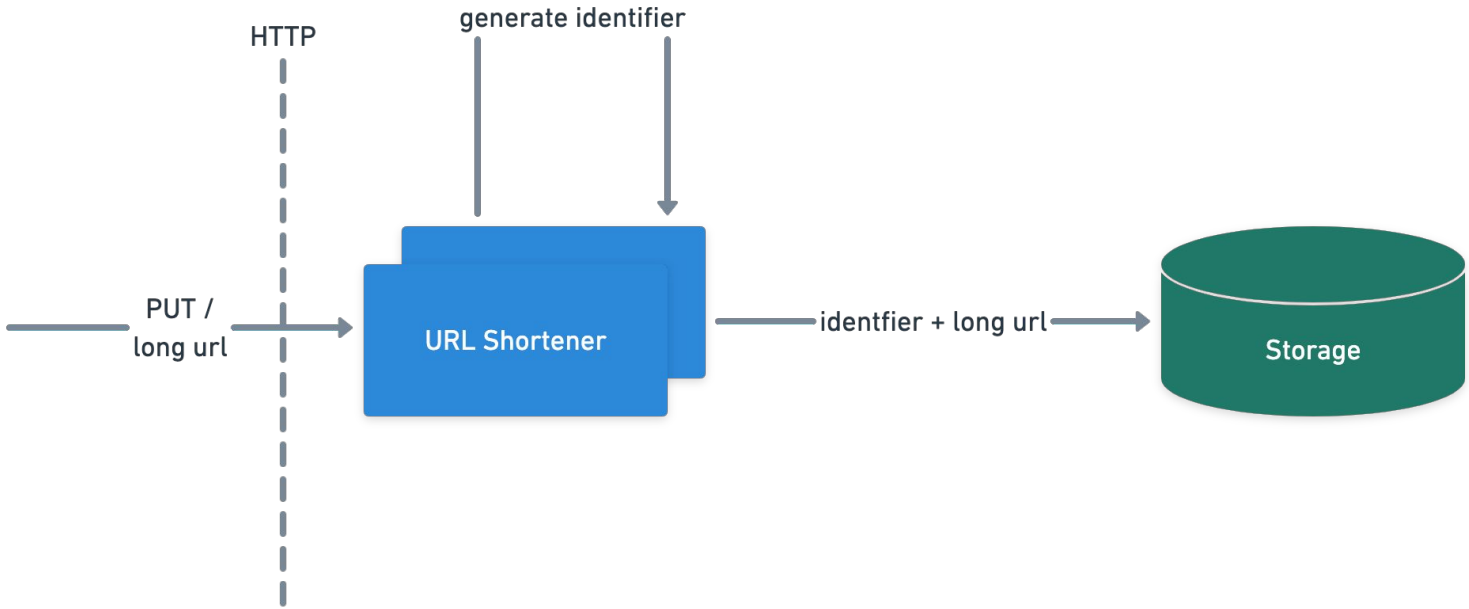
3 Billion KB =>

3 Million MB =>

3,000 GB =>

3 TB

Shortening URLs



Scaling reads

Read to write ratio

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300 writes per second.

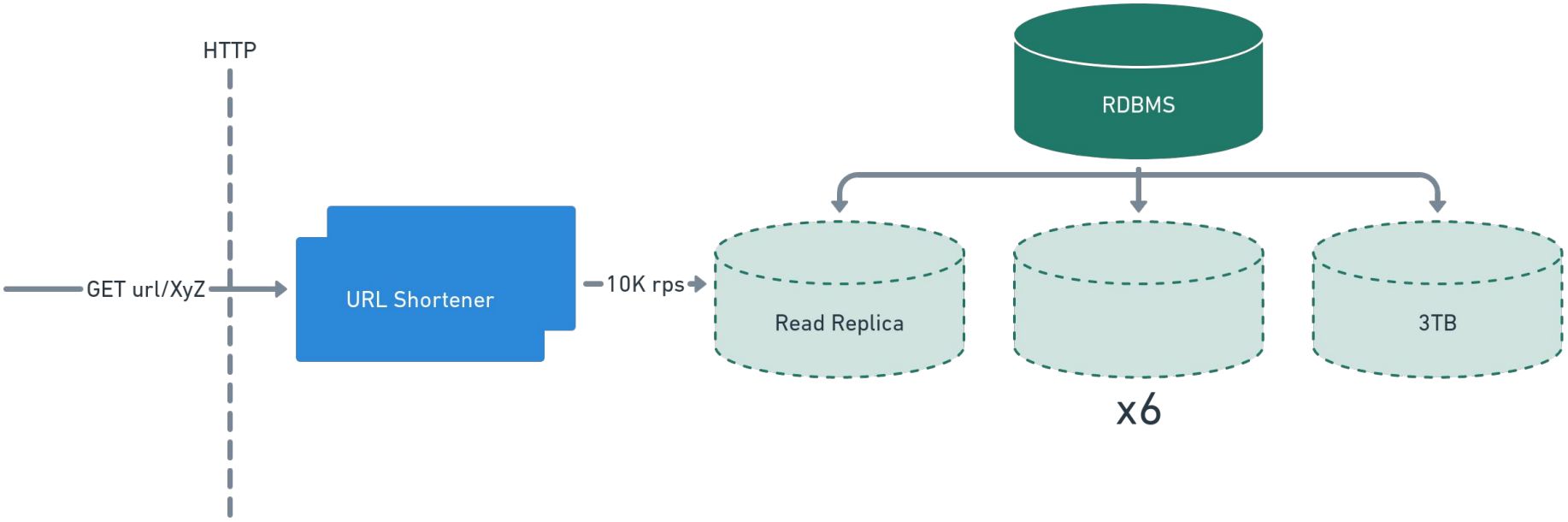
100:1 reads to writes =>

30K reads per second

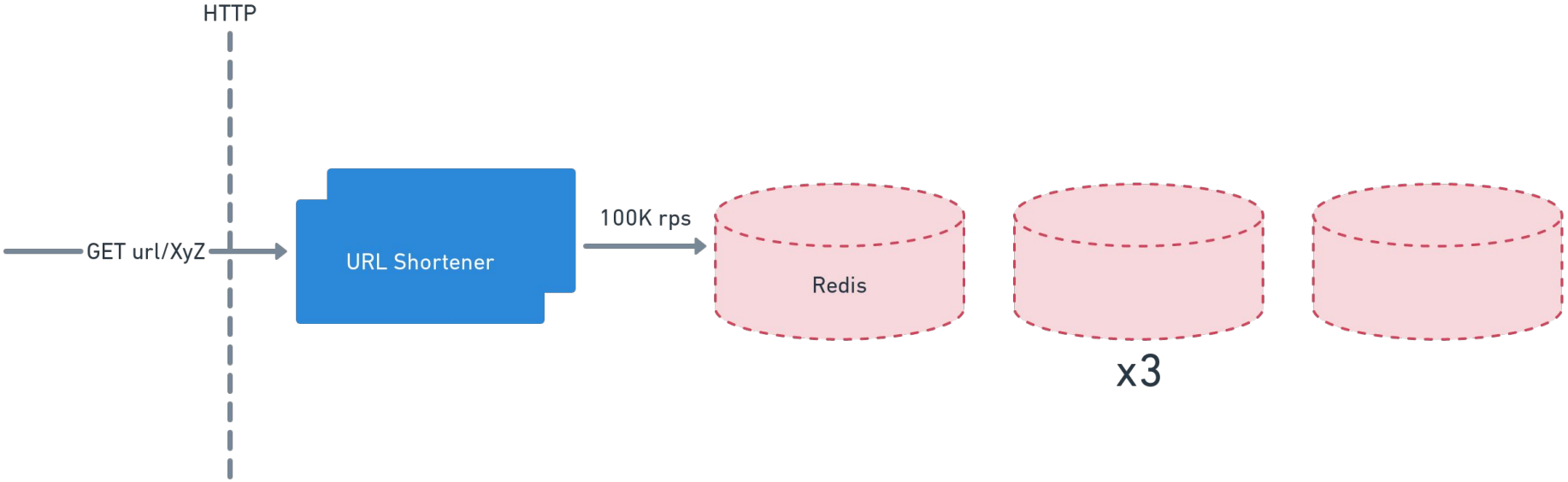
X10 peaks are possible

    300K rps on peak?

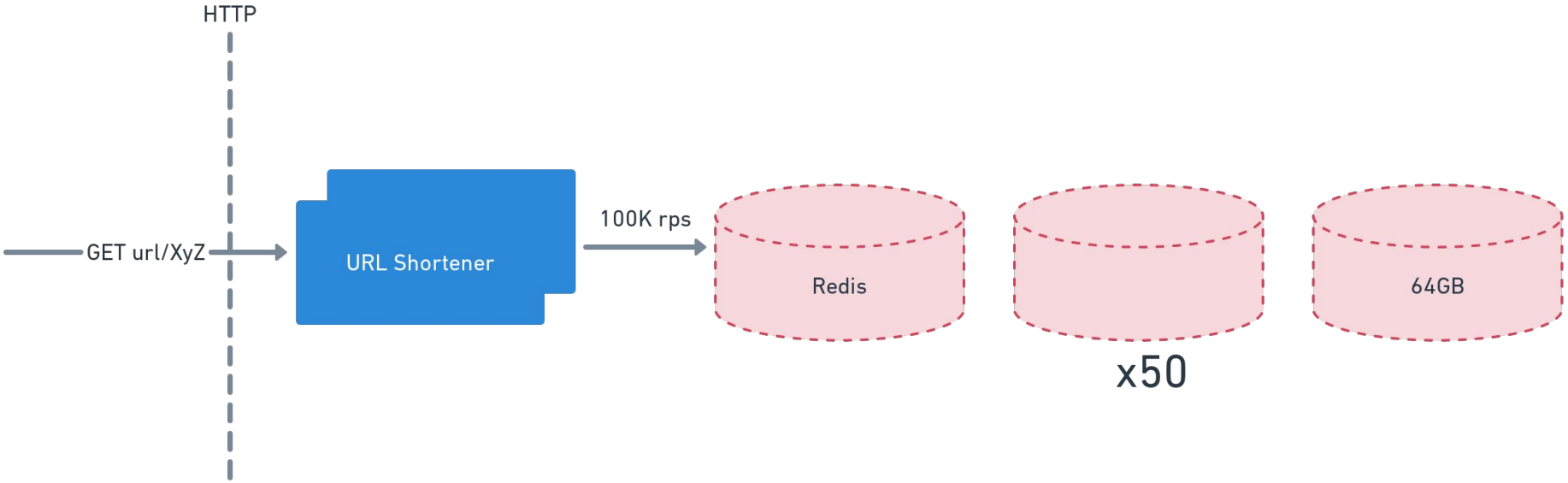
Scaling reads



Scaling reads

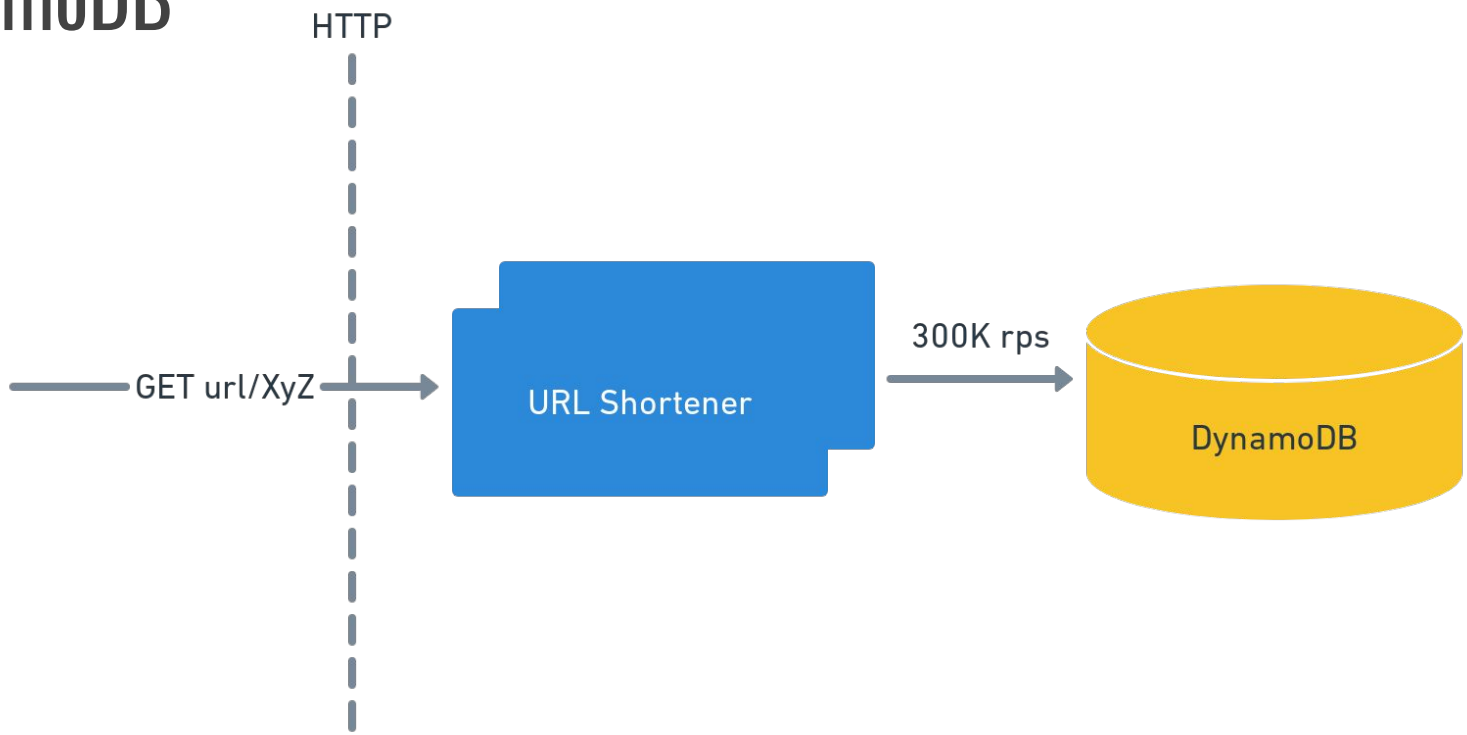


Scaling reads



DynamoDB

DynamoDB



Why not DynamoDB?

Pros:

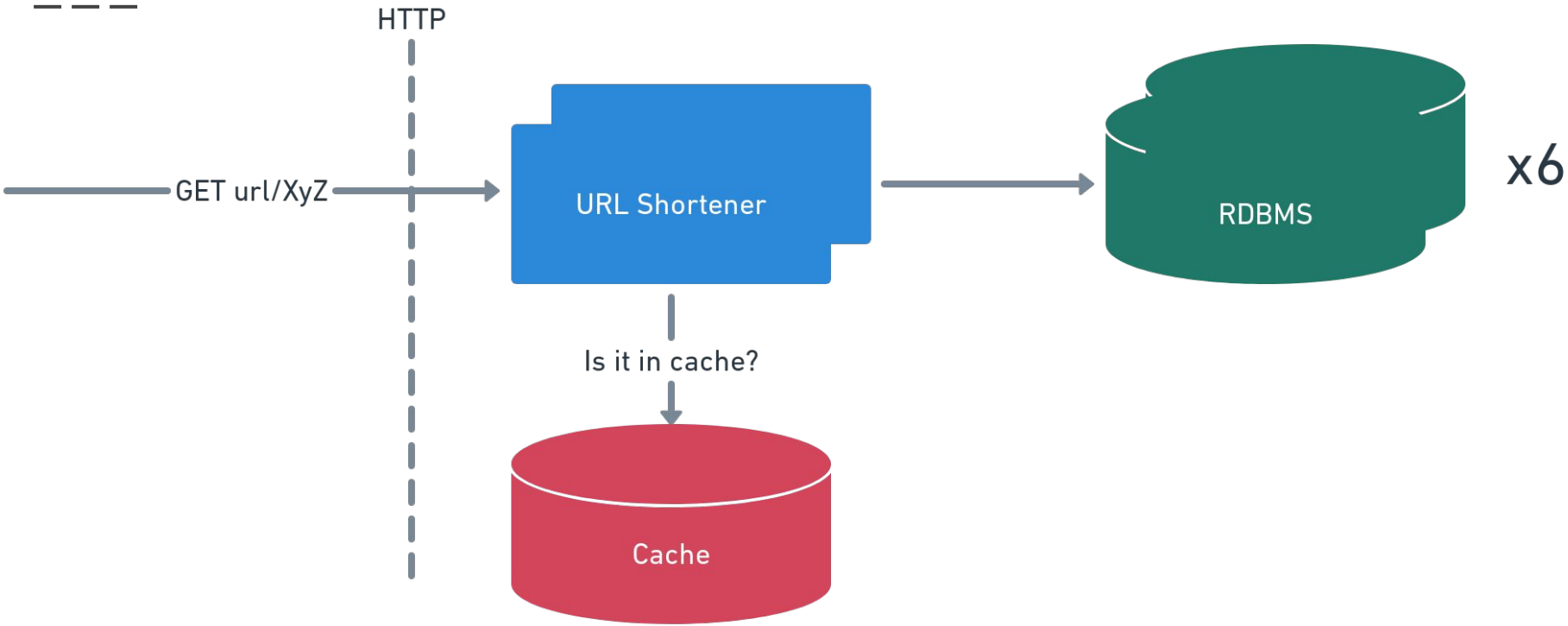
- Key-value store fits well with the design
- Sharded by default

Cons:

- Proprietary
- Price scales with cost

Caching

Caching



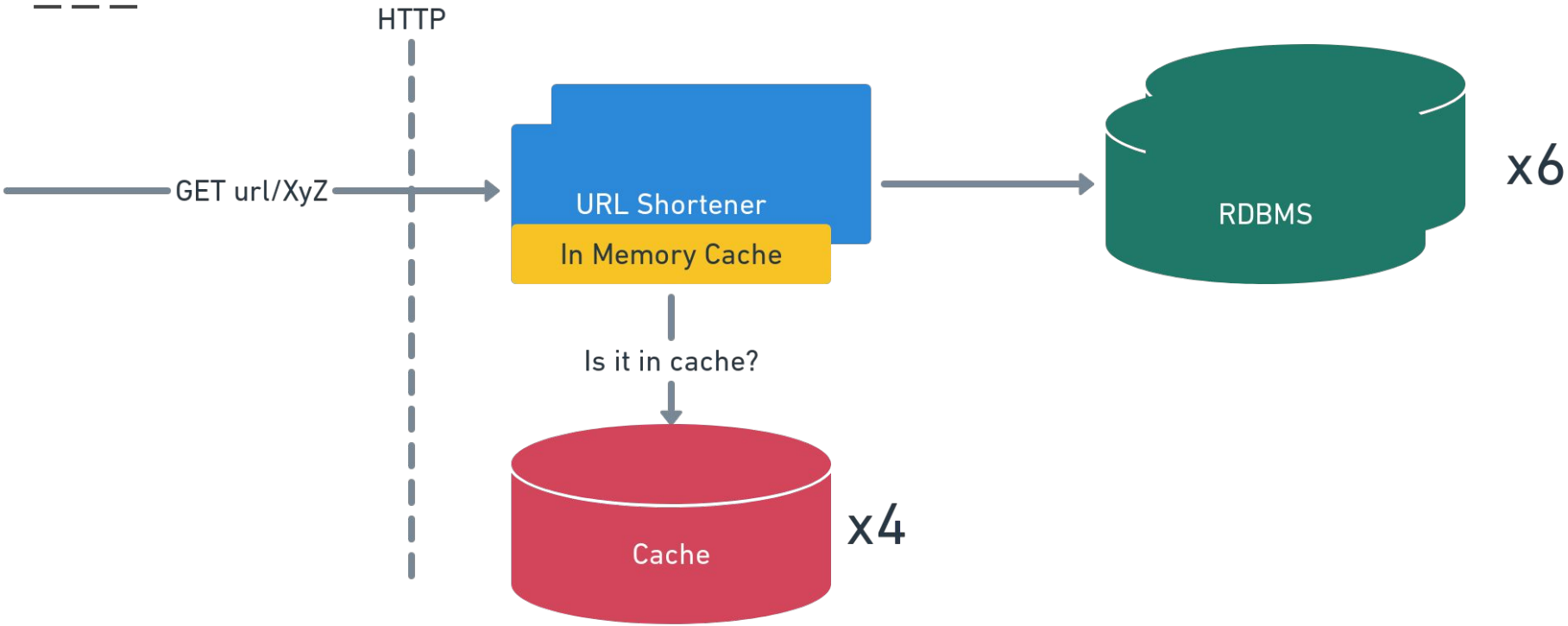
Caching

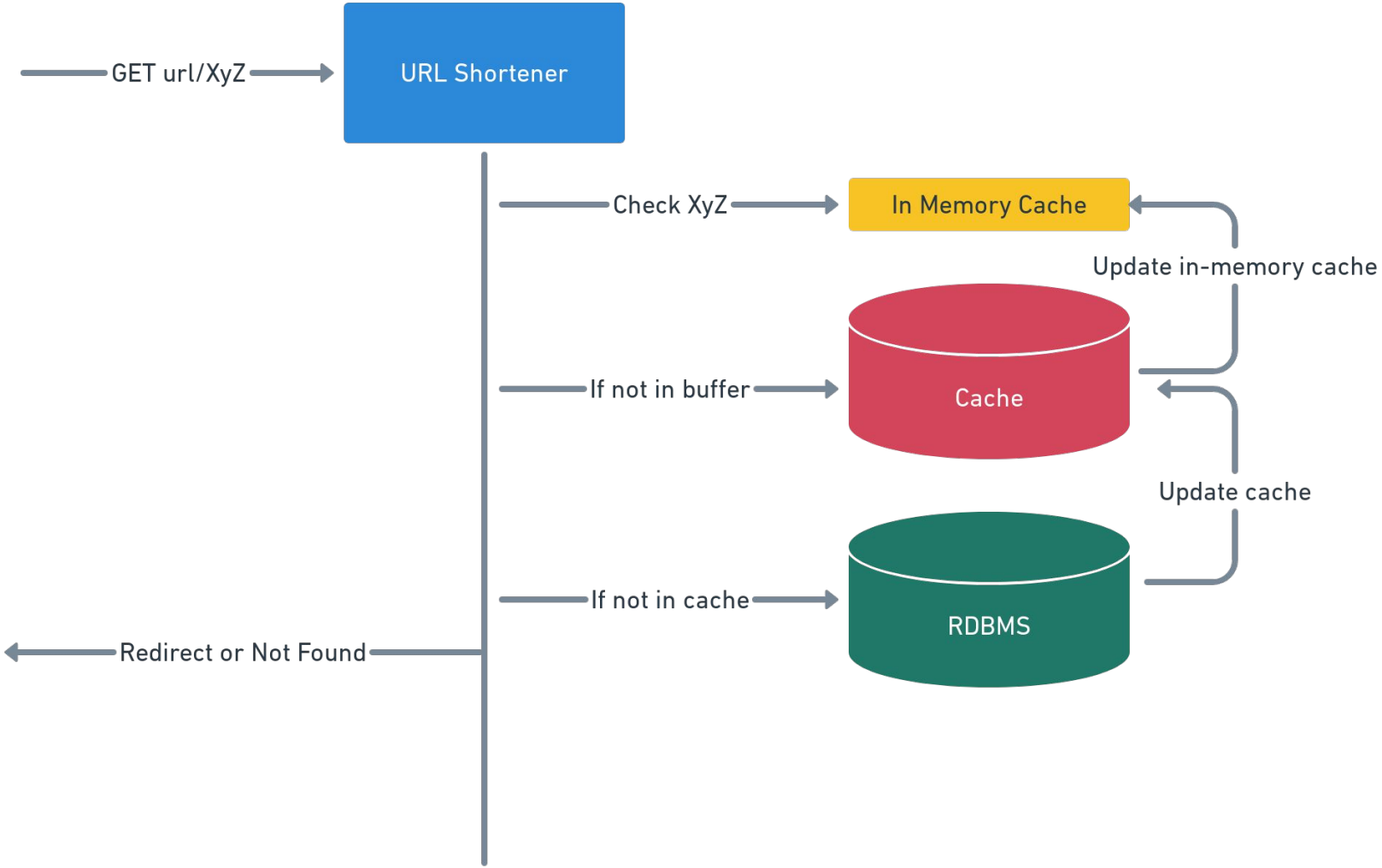
1% of 3TB => 30GB 😀

300K peak reads 😔

300K / 4 machines => less than 100K reads per second 😀

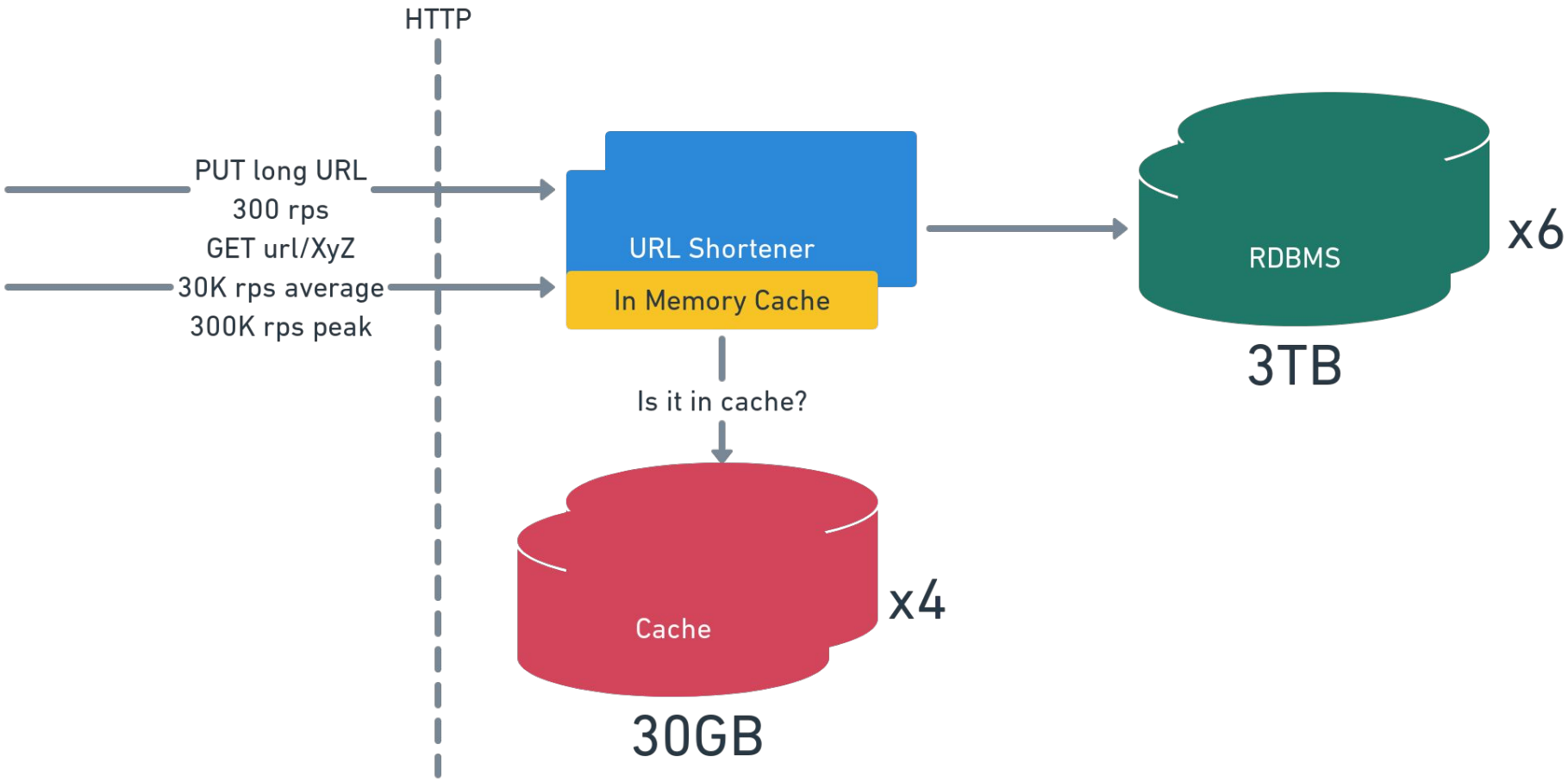
In memory caching




GET url/XyZ
URL Shortener
Check XyZ
In Memory Cache
Update in-memory cache
If not in buffer
Cache
Update cache
If not in cache
RDBMS
Redirect or Not Found

Summary

URL Shortener


HTTP
PUT long URL
300 rps
GET url/XyZ
30K rps average
300K rps peak
URL Shortener
In Memory Cache
RDBMS
x6
3TB
Is it in cache?
Cache
x4
30GB